

FPS Sensitivity Analysis: RoboReward-8B Performance Metrics

Task: Rearrange Pillows on Sofa | Dataset: DROID | Samples: 65 (50 Success, 15 Failure)

FPS	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Specificity (%)	TP	FN	FP	TN
default (2.0)	23.08	0.00	0.00	0.00	60.00	0	50	0	15
2.0	23.08	0.00	0.00	0.00	60.00	0	50	0	15
5.0	26.15	100.00	4.00	7.69	100.00	2	48	0	15
10.0	26.15	100.00	4.00	7.69	100.00	2	48	0	15
15.0	29.23	100.00	8.00	14.81	100.00	4	46	0	15
20.0	44.62	100.00	28.00	43.75	100.00	14	36	0	15
30.0	56.92	92.31	48.00	63.16	86.67	24	26	2	13
60.0 🏆	75.38	86.96	80.00	83.33	60.00	40	10	6	9

📊 Confusion Matrix Legend:
TP (True Positive): Model correctly predicts Success
FN (False Negative): Model misses Success (predicts Failure)
FP (False Positive): Model incorrectly predicts Success
TN (True Negative): Model correctly predicts Failure

🏆 Best Performance: 60 FPS achieves highest F1 Score (83.33%)